
SOFTWARE REQUIREMENTS SPECIFICATION

for

JANSAMVAAD

Unified Civic Issue Resolution Platform

Version 1.1

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1 Introduction

1.1 Background and Problem Statement

1.1.1 The Current Landscape

Urban environments generate thousands of civic issues daily, ranging from pothole-ridden streets and malfunctioning streetlights to unauthorized waste dumping. Currently, the mechanisms for reporting and resolving these grievances are highly fragmented. Citizens are forced to navigate a maze of disconnected web portals, physical municipal offices, and confusing helplines to get their voices heard.

1.1.2 The Challenges

- **High Friction and Low Engagement:** Existing civic apps often require mandatory downloads, tedious account creations, and steep learning curves. This high barrier to entry severely limits citizen participation.
- **The “Black Hole” of Reporting:** Once a report is submitted in legacy systems, there is virtually no visibility into its status. Citizens are left guessing whether their complaint was received, ignored, or acted upon, leading to systemic apathy.
- **False Closures and Lack of Accountability:** A critical flaw in current municipal workflows is the ability for officials to mark an issue as “resolved” without providing verifiable proof. This results in widespread false closures and heavily erodes public trust.
- **Manual and Error-Prone Routing:** Complaints are often manually sorted and forwarded to various departments. This analog approach results in bureaucratic delays, misclassified tickets, and highly inefficient resource allocation.

1.2 The JanSamvaad Solution

JanSamvaad is designed to dismantle these bureaucratic barriers by introducing a centralized, transparent, and highly automated framework for civic grievance redressal.

1.2.1 How It Solves the Problem

- **Zero-Friction Reporting:** By integrating a WhatsApp-based reporting interface alongside a robust web application, JanSamvaad eliminates the need for app downloads. Citizens can report issues instantly, with media attachments.
- **Intelligent Routing and Spatial Tagging:** Utilizing advanced spatial mapping for highly accurate location tagging and automated keyword analysis, the “Dispatcher Model” instantly routes the grievance to the correct jurisdictional department and field official. This completely removes human error and administrative bottlenecks from the sorting process.
- **Strict Verification Lifecycle & The Appeal Matrix:** JanSamvaad enforces a non-circumventable workflow that shifts the power of administrative closure to the citizen via a **7-day verification window**. To prevent fraudulent claims, the system utilizes **DigiPIN**

Lock a strict geospatial constraint requiring the citizen to be physically within 50 meters of the original issue to file an appeal. Unresolved issues trigger a two-tier escalation protocol directly to Department Admins, firmly combating the problem of false closures.

- **AI-Powered Moderation & System Integrity:** To maintain database integrity and prevent malicious flooding, an independent AI moderation layer acts as a gatekeeper before the routing engine. It enforces **rate limits** to prevent DoS attacks, performs **contextual validation** to ensure images match their descriptions, and executes a strict **“Three Strikes” Ban Protocol** that permanently revokes platform access for bad actors.
- **The Immutable Audit Trail:** The platform records a chronological, timestamped history of all state transitions, system assignments, uploaded proofs, and citizen appeals. This immutable ledger guarantees absolute transparency and prevents “he said, she said” disputes between citizens and municipal workers.
- **End-to-End Transparency:** Both the citizen and municipal stakeholders have real-time access to the exact status of the grievance. The Admin Dashboard provides data-driven insights to help officials track resolution efficiency, oversee appeals, monitor field worker performance, and identify recurring infrastructural failures in specific geographic zones.

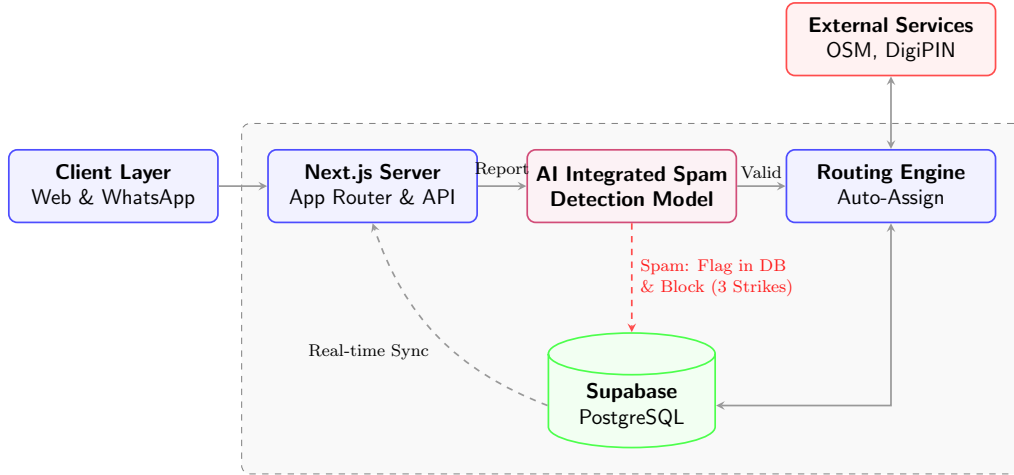


Figure 1.1: JanSamvaad High-Level System Architecture

2 Literature Review

The design of JanSamvaad is situated at the intersection of three evolving domains in civic technology: the shift toward “Address as a Service,” the affordances of instant messaging for low-literacy users, and the algorithmic management of public labor.

2.1 The Paradigm Shift in Location Identification

Traditional addressing in India has been described as a “barrier to digital transformation,” where the lack of standardized street names and numbers leads to massive operational inefficiencies in logistics and emergency response. Research into the DigiPIN protocol highlights that its strength lies in being a “pure function” of geography—it does not change even if the physical infrastructure is rebuilt. Literature suggests that such grid-based systems are essential for “Address as a Service” (AaaS), allowing municipal governments to provide standardized services to residents of informal settlements who were previously “invisible” to the administration.

2.2 WhatsApp and Participatory Instant Messaging (PIM)

The academic concept of “Participatory Instant Messaging” (PIM) explores how familiar platforms like WhatsApp can overcome the “app fatigue” that often kills e-participation initiatives. Studies show that for marginalized communities, including migrant workers and daily wage earners, a conversational interface is more trustworthy and accessible than a traditional web portal.

2.3 Dynamic Workload Management in Public Administration

Workload distribution is often the “Achilles heel” of public sector human resource management. Literature on Dynamic Workload Management Systems (DWMS) argues that manual assignment models often lead to “suboptimal resource allocation” and high levels of “employee exhaustion.” The “Dispatcher Model” utilized by JanSamvaad aligns with research proposing that task allocation should be based on objective “Capacity Factors” mathematical representations of an official’s current load and skill level rather than subjective administrative choices. This approach not only improves efficiency but also introduces a layer of “algorithmic fairness” into the municipal workplace.

2.4 Algorithmic Moderation and System Integrity

As civic technology platforms scale, they become vulnerable to data flooding and malicious submissions, which can paralyze municipal dispatchers. Recent literature on digital public infrastructure emphasizes the necessity of “Algorithmic Moderation” using AI models to validate the contextual relevance of citizen-submitted data before it enters the administrative workflow. By implementing automated spam mitigation and rate-limiting protocols, platforms can maintain database integrity, prevent denial-of-service (DoS) scenarios, and ensure that public sector resources are only allocated to genuine infrastructure failures.

3 Comparative Analysis: JanSamvaad vs. Legacy Solutions

To fully grasp the architectural and sociotechnical leap introduced by JanSamvaad, a rigorous comparison against established platforms like the Centralized Public Grievance Redress and Monitoring System (CPGRAMS) and the Swachhata App is required. This analysis extends beyond surface-level features to evaluate underlying infrastructure, state management, and user psychology.

Dimension	JanSamvaad	CPGRAMS	Swachhata App
Addressing & Mapping	Precision DigiPIN (4x4m grid)	Manual descriptive text input	Standard GPS drop-pins
Onboarding Interface	WhatsApp Chatbot (NLP driven gateway)	Complex Web Portal Forms	Dedicated Mobile App
Task Dispatcher	Algorithmic (Spatial + Capacity Factors)	Hierarchical / Manual forwarding	Automated regional push
State Machine	Strict 5-State Pipeline (with 7-Day Appeal Matrix)	Ambiguous (Open/Pending/Disposed)	3-State (Open/Resolved/Closed)
Verification Authority	Citizen-Led Validation & 2-Tier Appeal Protocol	Internal Administrative Review	Internal Official Photo Verification
Core Infrastructure	Next.js App Router, Supabase (PostgreSQL)	Monolithic, On-Premise Relational DBs	Hybrid Cloud Architecture
System Abuse Prevention	Contextual Moderation & 3-Strike Ban Protocol	Manual filtering & basic CAPTCHA	Basic IP rate limits
Transparency & Audit	Immutable Public Audit Timeline	Internal opaque backend logs	Limited static status updates

Table 3.1: Comprehensive Architectural and Functional Comparison

The primary differentiator for JanSamvaad is its focus on “resolution quality” over “disposal quantity.” Legacy systems often prioritize “closing” a ticket within the system, regardless of whether the physical problem was solved. JanSamvaad’s “Proof of Work” requirement coupled with the citizen-led 7-day verification window and strict geospatial appeal constraints ensures that accountability is built into the software logic itself, rather than relying on external administrative processes.

4 Overall Description

4.1 Product Perspective

JanSamvaad operates as a cloud-native solution. Unlike legacy systems, it introduces a **WhatsApp-First** approach to reduce entry barriers. It integrates with **DigiPIN** for location precision, utilizes an gated ingestion pipeline to protect database integrity, and enforces a closed-loop verification model backed by an immutable audit trail to ensure absolute accountability.

4.2 User Classes and Characteristics

1. Citizens:

- **Role:** Report issues, upvote nearby issues, verify “Proof of Work”, and initiate geospatial appeals if resolutions are unsatisfactory.
- **Characteristics:** Low technical expertise; reliant on mobile platforms (WhatsApp/Web).

2. Field Officials:

- **Role:** Receive algorithmically dispatched tasks, navigate to high-precision coordinates, upload verifiable resolution photos, and respond to Tier-1 appeals.
- **Characteristics:** Mobile-first usage in outdoor, potentially low-bandwidth environments.

3. Administrators:

- **Role:** Monitor zonal heatmaps, manage department definitions, adjudicate Tier-2 escalated appeals, and oversee AI-flagged user bans.
- **Characteristics:** High technical expertise; standard desktop usage.

4.3 Operating Environment

- **Frontend:** Next.js (App Router) for Server-Side Rendering (SSR) and responsive UI.
- **Messaging Gateway:** WhatsApp Business API via Webhooks.
- **Database & Auth:** Supabase (PostgreSQL + PostGIS for spatial queries, Row Level Security).
- **Mapping:** OpenStreetMap (OSM) integrated with the national DigiPIN grid.

5 System Features

5.1 Feature 1: Multi-Channel Onboarding (Web & WhatsApp)

To ensure maximum accessibility across diverse demographics, citizens can report issues either through a dedicated, responsive web portal or a frictionless WhatsApp chatbot gateway. The platform acts as a unified funnel, accepting media and location data from multiple sources and normalizing it before it hits the central dispatcher.

5.2 Feature 2: Operational Workflow & The Audit Trail

The system enforces a strict state-machine workflow designed to shift closure authority from the administration to the citizen. The core lifecycle moves through a streamlined pipeline: **Active** → **Under Progress** → **Under Review** → **Closed**.

To combat false closures, transitioning an issue to “Closed” triggers a 7-day verification window. During this window, citizens can evaluate the official’s photographic proof and initiate a structured Appeal Protocol (moving the state to **Under Review**). This appeal process is governed by a strict geospatial “DigiPIN Lock” to prevent fraudulent claims, and features a two-tier escalation path up to Department Administrators. An immutable audit trail transparently logs every interaction within this lifecycle.

5.3 Feature 3: Data-Driven Dispatcher Model

The system assigns tasks based on objective data points (Spatial Proximity + Workload Capacity) rather than subjective manual selection, ensuring equitable labor distribution.

Unlike legacy manual dispatching, the JanSamvaad routing engine acts as a fully automated orchestration layer. It bridges the AI moderation ingestion pipeline, external geospatial APIs (OpenStreetMap/DigiPIN), and internal database state management to resolve assignments in milliseconds.

5.4 Feature 4: Administrative Analytics & Oversight

The administrative dashboard provides specialized modules for high-level oversight. This includes dynamic spatial heatmaps to identify recurring infrastructural failures, a moderation suite to oversee AI-flagged spam bans, and verification auditing tools to track official performance metrics across the appeal lifecycle.

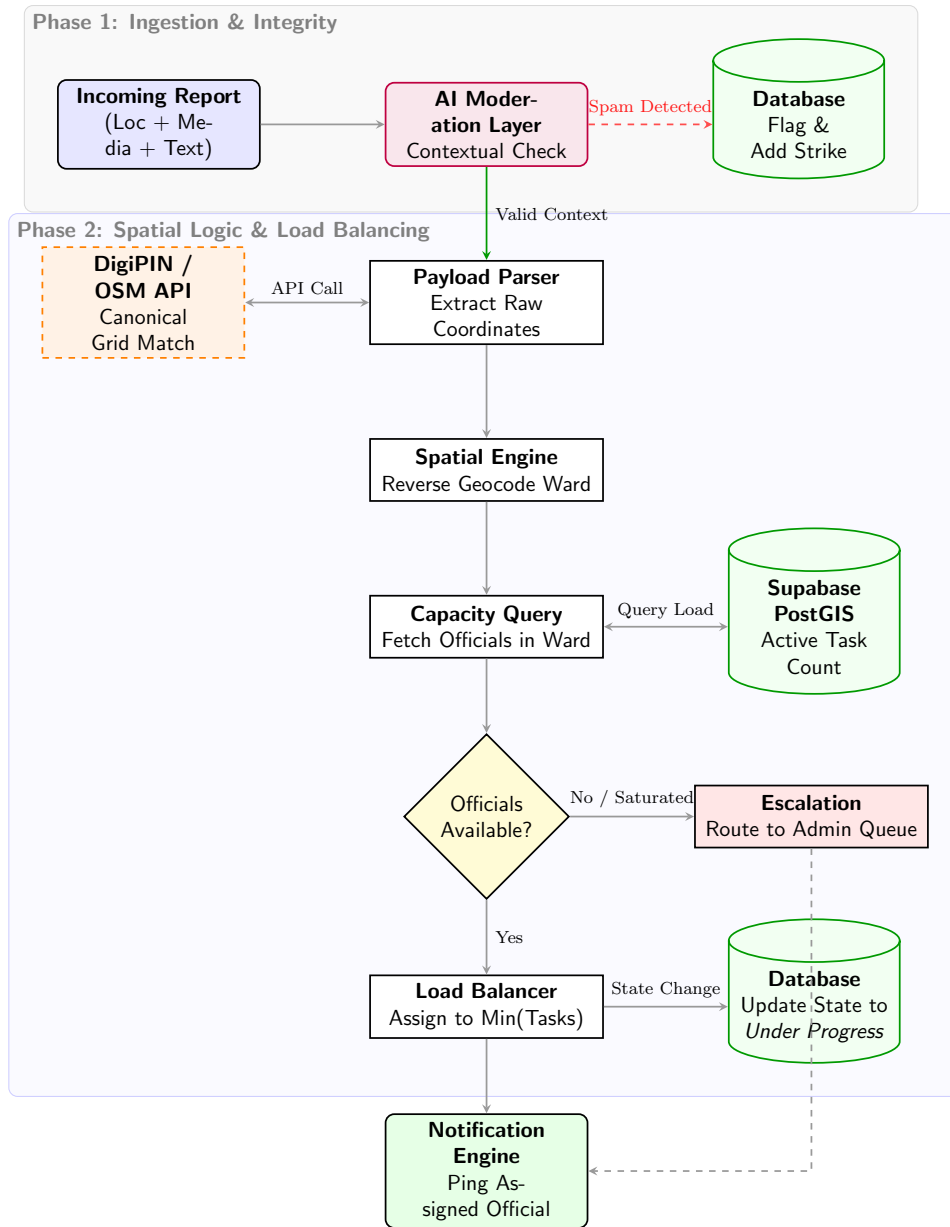
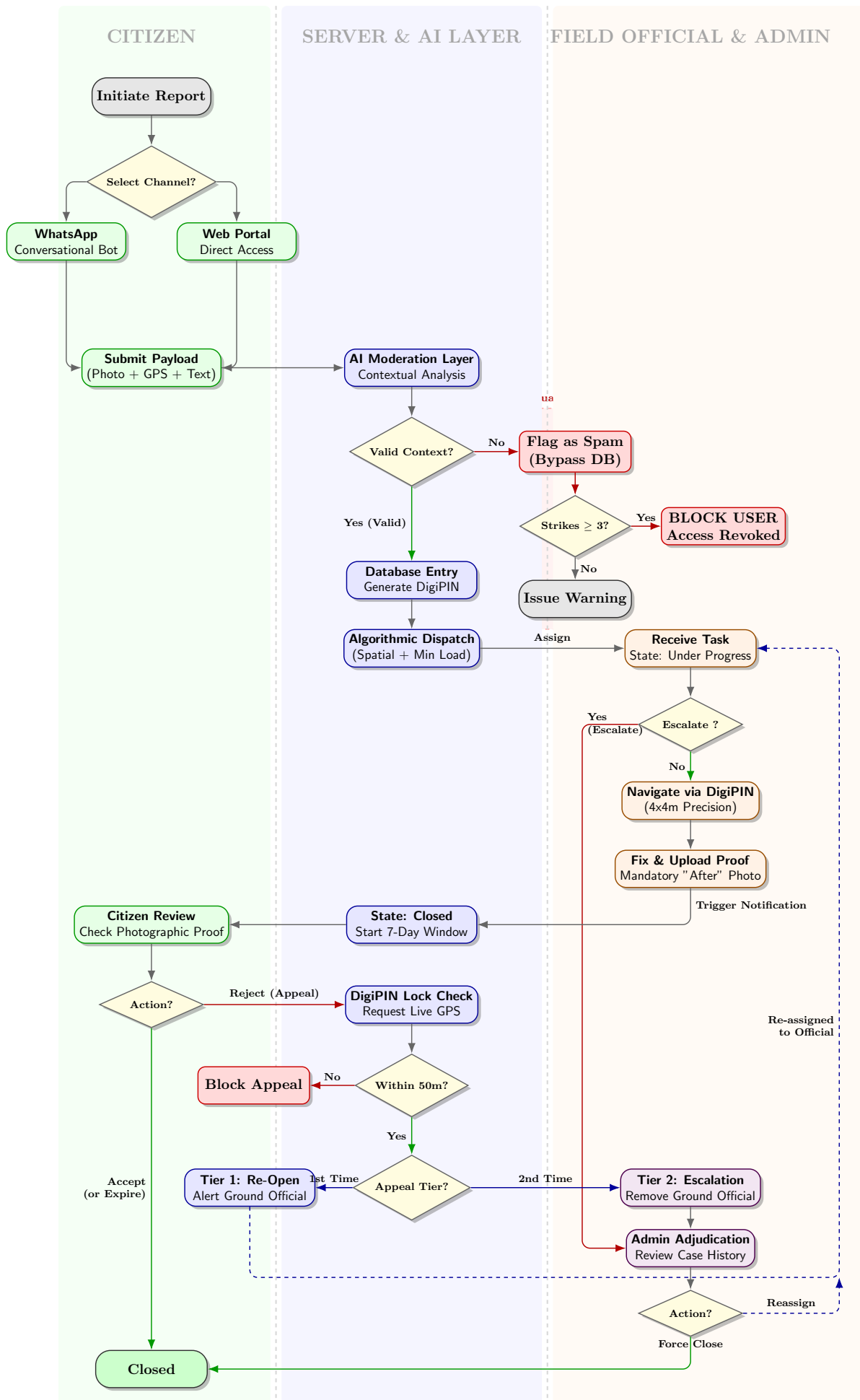


Figure 5.1: End-to-End Orchestration of an issue: From Ingestion to Load-Balanced Dispatch



6 Functional Requirements

The functionality of JanSamvaad is governed by a set of rigorous requirements designed to ensure that the transition from grievance to resolution is transparent, data-driven, and verifiable.

6.1 Onboarding and Submission

To maximize accessibility, the system must support two primary channels for report submission: a dedicated web portal and a WhatsApp chatbot.

- **WhatsApp Bot Interaction:** The system shall accept an initial stimulus (e.g., a "Hi" message or a photo) via WhatsApp. The bot shall navigate the citizen to a structured reporting flow.
- **Web Portal Accessibility:** The Next.js web portal shall provide a responsive interface for detailed logging, allowing users to submit textual descriptions, category tags, and photographic evidence.
- **Automated Geotagging:** At the moment of submission, the platform must automatically capture GPS coordinates and convert them into a DigiPIN code. This process must be transparent to the user, ensuring location precision without manual input.

6.2 Operational Workflow and Lifecycle Management

The system enforces a strict state-machine workflow to prevent the premature or fraudulent closure of civic issues. The issue lifecycle follows a non-linear but strictly defined pipeline:

- **State 1: Active:** The issue is verified by the system logic and enters the dispatcher queue. It is visible to nearby citizens for upvoting.
- **State 2: Under Progress:** An official has been assigned the task to resolve the issue. The system notifies the original reporter that work has commenced, building institutional trust.
- **State 3: Closed:** The official has completed the physical task. To trigger this state, the official is required to upload an "After" photo as Proof of Work. This is a mandatory constraint that cannot be bypassed. Once uploaded, the system notifies the original reporter and top upvoters to verify the resolution through an "Appeal" interaction.
- **State 4: Under Review (Rejection Loop):** If the citizen rejects the proof, indicating the issue is not genuinely resolved, the issue enters a "Appeal Loop." The state is reverted from "Closed" to "Under Review" for administrative assessment and potential escalation.
- **State 5: Closed:** If the citizen accepts the proof and verifies that the issue is solved, the issue is permanently marked as "Closed" completing the verification lifecycle.

7 Non-Functional Requirements and System Constraints

The resilience of JanSamvaad depends on non-functional parameters that ensure performance, security, and privacy across the municipal landscape.

7.1 Performance and Scalability

- **Real-time Updates:** The backend must support real-time data persistence through Supabase, ensuring that status changes are reflected on the citizen dashboard and administrative heatmaps with sub-second latency.
- **Geospatial Concurrency:** The system must handle high-volume spatial queries, particularly the cross-referencing of new reports against existing ones to detect duplicates within a specific spatial radius (Nearby Issues).

7.2 Security and Role-Based Access Control (RBAC)

- **Database-Level Enforcement:** Access rights for Citizens, Officials, and Admins must be strictly enforced at the database level using Row-Level Security (RLS) to prevent unauthorized data manipulation.
- **Authentication Protocols:** Official access must be gated through verified, domain-specific authentication (e.g., government-issued email or mobile verification) to ensure that only authorized personnel can update the status of infrastructure repairs.

7.3 Data Integrity and Privacy (DPDP Compliance)

- **Secure Backend Proxying:** The system must use secure proxying for external services (like mapping APIs) to protect sensitive coordinate data and adhere to Content Security Policies (CSP).
- **Compliance with DPDP Act 2023:** As the system handles personal data (locations & photos), it must implement a "Consent-by-Design" framework. Citizens must be provided with clear notices regarding data usage, and the system must support the right to erasure and correction as mandated by the Digital Personal Data Protection Act.

8 Use Case Scenarios

The following scenarios illustrate the end-to-end operational flow of JanSamvaad across its diverse user base.

8.1 Use Case 1: The Citizen - Reporting a Streetlight Failure

- **Identification:** A citizen notices a non-functional streetlight.
- **Navigation:** The WhatsApp bot serves as a gateway, providing a direct link to the web portal.
- **Submission:** The citizen accesses the portal, uploading a photo and a short description.
- **Location:** GPS coordinates are captured and encoded into a 10-character DigiPIN, creating an “Active” ticket.
- **Community Engagement:** Neighbors can upvote, increasing the issue’s priority on the administrative heatmap.
- **Resolution Loop:** Upon repair, the citizen reviews the “After” photo.

8.2 Use Case 2: The Official - Executing a Pothole Repair

- **Notification:** A field official receives a new task assignment in their ward.
- **Navigation:** The official uses the DigiPIN link to open a high-precision map (4m x 4m grid), navigating directly to the site without needing manual directions.
- **Status Update:** The official updates the status to “Under Progress.”
- **Verification:** Post-repair, the official uploads an “After” photo. The system verifies the spatial and temporal metadata.
- **Completion:** The official updates the status to “Closed,” triggering a completion notification to the citizen.

8.3 Use Case 3: The Administrator - Analyzing Systemic Bottlenecks

- **Strategic View:** An administrator reviews “Response Time” metrics on the Next.js dashboard.
- **Insight:** Statistics reveal a high density of unresolved reports in the Eastern Zone.
- **Action:** Workload reports show local officials are at “saturation level” (exceeding 10 active tasks).
- **Optimization:** The administrator temporarily reassigns a low-workload official from a neighboring zone to assist.

9 Concurrency with Government Policies and Vision

JanSamvaad is architected not merely as a grievance redressal tool, but as a Digital Public Good (DPG) aligned with the Government of India's vision for "Viksit Bharat" (Developed India). Much like *DigiYatra* served as a tested, mid-level application proving the efficacy of biometric integration before national scaling, JanSamvaad serves as a foundational proof-of-concept for hyper-local, geospatial governance.

9.1 Alignment with National Geospatial Policy (NGP) 2022

The NGP 2022 emphasizes "Atmanirbhar Bharat" in geospatial technologies, urging a move away from dependency on foreign proprietary mapping solutions toward indigenous standards.

- **Adoption of DigiPIN:** By utilizing the Department of Posts' Digital Postal Index Number (DigiPIN), JanSamvaad adheres to the NGP's mandate for a standardized National Address Grid. This ensures that every grievance reported is tied to a canonical, government-recognized location code rather than arbitrary coordinates.
- **Open Standards Interoperability:** The system's integration with OpenStreetMap (OSM) and standard geospatial queries fulfills the policy's requirement for open data interchange, ensuring that data generated by JanSamvaad is not siloed but is interoperable with national frameworks like DHRUVA.

9.2 Adherence to the DPDP Act 2023

With the passing of the Digital Personal Data Protection (DPDP) Act, 2023, the handling of citizen data requires strict architectural safeguards.

- **Privacy by Design:** The implementation of Row-Level Security (RLS) at the database level ensures that data access is restricted based on strict necessity. This aligns with the Act's principle of "Purpose Limitation."
- **Secure Proxying:** By proxying external mapping services, JanSamvaad minimizes the exposure of user metadata, adhering to the "Data Minimization" principles required for trusted digital ecosystems.

9.3 Smart Cities Mission: Data-Driven Governance

The Smart Cities Mission relies on data harvesting to improve urban infrastructure.

- **Geospatial Heatmaps:** The system's ability to generate heatmaps from aggregate report data transforms reactive grievance redressal into proactive urban planning. This provides municipal bodies with the "Evidence-Based Decision Making" capabilities envisioned by the Smart Cities framework.

10 Conclusion

10.1 Bridging the Digital and Spatial Divide

JanSamvaad emerges as a necessary evolution in municipal governance, directly addressing the systemic inefficiencies that have long plagued civic grievance redressal in India. By fundamentally rethinking how citizens interact with the state, the platform dismantles the traditional barriers of "app fatigue" and digital illiteracy. The integration of a WhatsApp-first conversational interface ensures that the system is accessible to the widest possible demographic, meeting the user where they already are. Concurrently, the implementation of the DigiPIN protocol resolves the chronic issue of location ambiguity. By transforming arbitrary geographic descriptions into a standardized, canonical 10-character code mapped to a 4x4 meter grid, JanSamvaad ensures that every reported issue is pinned with absolute precision. Together, these two pillars transform an historically opaque reporting process into a frictionless, universally accessible digital experience.

10.2 Enforcing Algorithmic Accountability and Transparency

The core innovation of JanSamvaad lies in its ability to embed accountability directly into its software architecture, rather than relying on external administrative oversight. Traditional legacy systems, such as CPGRAMS, often prioritize the mere disposal of tickets, leading to premature or fraudulent closures. JanSamvaad counteracts this through its rigorous lifecycle and the mandatory "Proof of Work" protocol. An issue cannot be marked as "Closed" without photographic evidence from the field official and, crucially, a final binary validation from the reporting citizen. If not explicitly verified by the user, it remains strictly in a "Closed" state pending the 7-day verification window. Furthermore, the Data-Driven Dispatcher eliminates manual bias in task assignment, utilizing spatial awareness and capacity factors to distribute workloads equitably. This algorithmic approach not only optimizes municipal resources but establishes a transparent, tamper-proof ecosystem where actual physical resolution is the only metric for success.

10.3 A Foundation for Data-Driven Digital Governance

Ultimately, JanSamvaad transcends its primary function as a grievance redressal tool to establish a scalable framework for a Digital Public Good. By strictly adhering to the mandates of the Digital Personal Data Protection (DPDP) Act 2023 and the National Geospatial Policy (NGP) 2022, the platform secures citizen privacy while championing indigenous, open-standard geospatial technologies like OpenStreetMap. The transition from reactive management to proactive urban planning is realized through administrative heatmaps and AI-driven spam mitigation, which together protect system integrity and highlight systemic infrastructure bottlenecks. In conclusion, JanSamvaad successfully proves that hyper-local, decentralized governance is achievable at scale, restoring public trust and laying a robust, data-driven foundation for the future of India's Smart Cities.

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